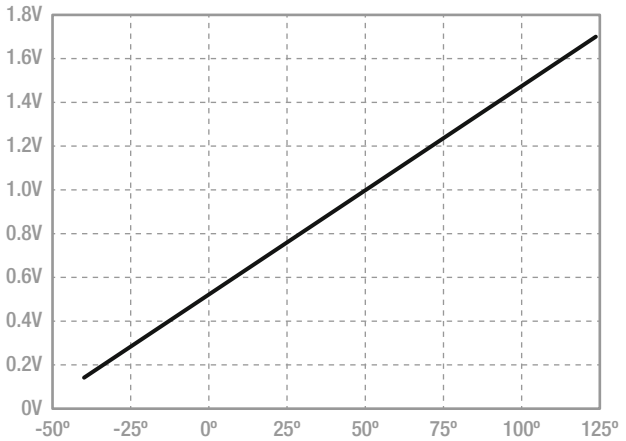


Figure 7-2. *TMP36 output voltage versus temperature Celsius*

The TMP36 has a 0.5v offset that we will have to compensate for, but because of the linearity of the output signal only a little math is needed to determine the ambient temperature to within ± 2° Celsius. There are many other temperature sensors like these that we could use or we could add additional sensors, like ones for measuring humidity or barometric pressure, but we are going to keep it simple for this project. You might also want a little more power out of your LEDs, so you could substitute one of the other varieties of the BlinkM, even connecting a MaxM to an entire string of LEDs. For more information and to download the data sheet and quick-start guide, check out ThingM's web site at `http://thingm.com/products/blinkm/`. Now, let's connect our project, as shown in Figures 7-3 and 7-4.

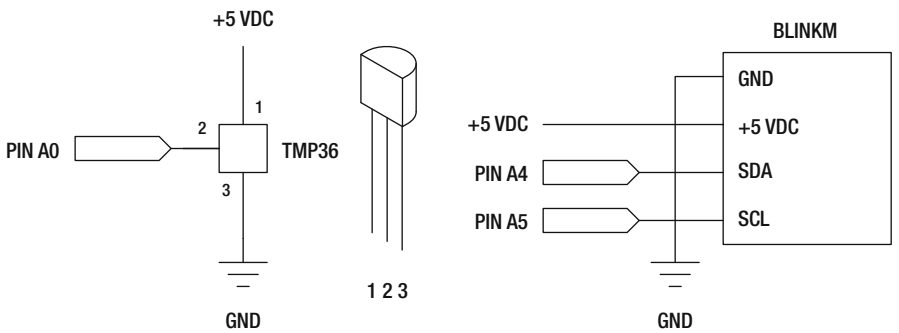


Figure 7-3. *Ambient Temps schematic*